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# ECE 6110 - Lab Assignment 1
# Topologies, Logging, and PCAP Tracing

## Part 1b

Now, we use the ping application instead of the echo one. We can again measure the round-trip delay with a delay of 5Mbps.

| nRelays | dataRate (Mbps) | Measured Round-trip Delay (s) with Ping |
|---|---|---|
| 1 | 5 | 0.008551 |
| 1 | 10 | 0.008275 |
| 2 | 5 | 0.012826 |
| 2 | 10 | 0.012413 |
| 3 | 5 | 0.017101 |
| 3 | 10 | 0.016551 |
| 4 | 5 | 0.021376 |
| 4 | 10 | 0.020688 |
| 5 | 5 | 0.025651 |
| 5 | 10 | 0.024826 |

We observe that the measured round-trip delay with the ping protocol is faster than the round-trip delay using an echo application. This is because the ping uses a layer 3 protocol, the Internet Control Message Protocol (ICMP), while the packets sent thanks to the echo application use the User Data Protocol (UDP), a layer 4 protocol. As a result, a ping packet has fewer headers to handle than echoed packets. Moreover, ping packets are much smaller than the ones sent with UDP.